Midterm Mock Exam (Practice Version)

Instructions

This practice exam has the same format as the real midterm: four parts, 100 points, 100 minutes. The scenario and the numbers are different, so memorizing answers will not help. Try it alone under time pressure, then compare with the solutions your instructor releases.

- **Covers:** Chapters 1 to 4 (Session 1): decision judgment, leakage, bias and variance, regression, logistic regression, the confusion matrix, and cost and threshold.
- **Allowed:** a calculator and one A4 sheet of your own notes. No AI tools and no laptops in the real exam.
- **How to answer:** show your working. A formula sheet is printed with the exam, so you do not need to memorize formulas. A correct number with no working earns partial credit at best. Short answers should be one to three sentences.

Scenario for Parts B to D. A hospital wants to phone patients after discharge to prevent a readmission within 30 days. A phone call is the action. A missed readmission (a patient who needed a call but did not get one and came back) costs 9,000. An unnecessary call costs 1,000. All money is in fictional units.

Formula sheet (printed with the exam). "Flagged" means the model recommends the action. TP: flagged and the event happened. FN: not flagged but the event happened. FP: flagged but it did not happen. TN: not flagged and it did not happen.

Quantity	Formula
Accuracy	(TP + TN) / (TP + TN + FP + FN)
Sensitivity (recall)	TP / (TP + FN)
Specificity	TN / (TN + FP)
Precision	TP / (TP + FP)
Total cost	C_FN × FN + C_FP × FP
Break-even probability	p* = C_FP / (C_FP + C_FN)
Logit to probability	z = b0 + b1·x1 + b2·x2, then p = e^z / (1 + e^z)
Odds and odds ratio	odds = p / (1 − p); odds ratio = e^b
Linear prediction	ŷ = b0 + b1·x1 + b2·x2

Part A. Concept checks (12 points: items 1 to 8 are multiple choice, 1 point each; items 9 and 10 are true or false with a one-sentence reason, 2 points each)

1. The hospital adds the field "number of follow-up calls made after discharge", and test accuracy rises from 88% to 99%. What should the hospital do?
   (A) Deploy the model: 99% is excellent.
   (B) Check when each input becomes known, and drop any input that is not available at discharge.
   (C) Add more fields like it.
   (D) Lower the threshold.
2. A team tries ten models and reports the best test accuracy as what the hospital should expect. What is the right fix?
   (A) Report training accuracy instead.
   (B) Choose the model on a validation set (or by cross-validation) and use the test set once, at the end.
   (C) Pick the model with the most inputs.
   (D) Use a larger test set only.
3. Readmissions are recorded only for patients who attended the follow-up clinic, and the team wants to score all discharged patients. What is the best first step?

(A) Treat patients who did not attend as not readmitted.
(B) Treat patients who did not attend as readmitted.
(C) Compare attenders with all discharged patients, and follow a small random sample of non-attenders to learn about the rest.
(D) Drop the non-attenders and continue.

4. A team lowers K in K-nearest neighbors from 25 to 1. Training error falls to zero and validation error rises. What should the team do?
(A) Keep K = 1, because training error is zero.
(B) Raise K and choose it by validation error: the model has become too flexible (low bias, high variance).
(C) Lower K further.
(D) Drop the validation set.

5. Adding 20 more inputs raises R^2 on the training data from 0.62 to 0.71, but R^2 on the test data falls from 0.60 to 0.55. What does this show?
(A) The new inputs help.
(B) Training R^2 never falls when inputs are added, so it proves nothing; the falling test R^2 shows overfitting.
(C) R^2 is useless for any model.
(D) The test set is wrong.

6. Two readmission models have the same test performance. One uses 8 inputs and the other 80. Which do you choose for the hospital?
(A) The 80-input model, because more inputs means more information.
(B) The 8-input model: easier to explain and audit, less likely to overfit, cheaper to monitor, unless the 80-input model clearly lowers cost.
(C) Either one: performance is the same.
(D) The 80-input model, because its training accuracy is higher.

7. A vendor says its AI tool is “95% accurate”. Only 15% of patients are readmitted. What is the best response?
(A) Buy it.
(B) Ask how a rule that calls nobody would score, and what each error costs.
(C) Ask for its training accuracy.
(D) Ask how many inputs it uses.

8. Which is the best evidence that a model will work on next quarter’s patients?
(A) High accuracy on the training data.
(B) High accuracy on a random split of the same file.
(C) Good results on recent patients the model has never seen, including groups that are missing from the training file.
(D) A large number of inputs.

9. True or false, with one sentence: a more flexible model always predicts new cases better than a simpler one.

10. True or false, with one sentence: a model with 95% accuracy is always better for the hospital than a model with 90% accuracy.

Part B. Reading and comparing regression models (34 points)

A logistic regression predicts whether a patient is readmitted within 30 days.

Term	Coefficient	Std. error	p-value
Intercept	-2.20	0.15	< 0.001
age_over_75 [Yes]	0.69	0.25	0.006

Term	Coefficient	Std. error	p-value
prior_admissions (per admission)	0.40	0.08	< 0.001

1. (3 points) Compute the predicted probability of readmission for a patient over 75 with two prior admissions. Show your working.
2. (5 points) (a) Interpret the coefficient 0.40 for prior admissions in terms of odds, and state what is held fixed. (b) A colleague says: "age_over_75 (0.69) matters more than prior_admissions (0.40) because its coefficient is bigger." Explain why that comparison is not fair.
3. (6 points) A manager says: "prior_admissions has $p < 0.001$, so phoning patients with many prior admissions will reduce their readmissions." (a) Is this claim supported? Explain. (b) Describe an A/B test that could support it: who is randomly assigned, what is compared, and why does random assignment matter? (c) Name one thing you would check before trusting the result.
4. (4 points) The training data contain only patients who were discharged home, never patients sent to a nursing facility. What does this mean for scoring a patient who is about to be sent to a nursing facility? What would you do?
5. (4 points) A linear regression fitted on patients aged 60 to 95 gives: length of stay (days) = 2.1 + 0.05 × age, with training $R^2 = 0.12$ and residual standard error 3.4 days. (a) Interpret the coefficient 0.05 in words, with units. (b) The first table below shows the residuals by age band. What do you conclude about using a straight line here, and about how precise the predictions are for older patients?
6. (4 points) (a) In Model 3 (second table below) neither coefficient is significant, although each input was significant alone. Why? (b) Can you say that one more chronic condition adds 0.25 days with age held fixed? (c) What would you do if the goal is prediction, and what if the goal is explanation?
7. (4 points) The team built three logistic regressions (third table below). (a) Which model overfits, and how can you tell? (b) Which model do you recommend to the hospital, and why? Consider test performance and how easily clinicians can understand it. (c) Name one thing you would check before paying for the extra data behind Model 2.
8. (4 points) In the logistic regression at the top of this part, the coefficient for age_over_75 is 0.69 with standard error 0.25. (a) Give an approximate 95% confidence interval (the coefficient plus or minus two standard errors) and say in words what it means. (b) The p-value is 0.006. What does the interval tell you that the p-value does not? (c) A team copies every row of the data twice and refits the model. The standard error shrinks. Is the estimate now more reliable? Why or why not?

Table for item 5. Residuals of the length-of-stay line (actual minus predicted, in days), by age band.

Age band	Patients	Mean residual	Spread (standard deviation)
60 to 69	120	-1.1	2.0
70 to 79	200	-0.2	3.0
80 to 89	140	+0.6	4.2
90 and over	40	+1.5	5.5

Table for item 6. Three linear regressions for length of stay (days). Age and chronic conditions are strongly related (correlation 0.90).

Model	Inputs	Coefficients (standard error)	R^2
1	age	age 0.05 (0.01)	0.12
2	chronic_conditions	conditions 0.42 (0.07)	0.13
3	age and chronic_conditions	age 0.02 (0.02); conditions 0.25 (0.16)	0.14

Table for item 7. AUC is the area under the ROC curve: higher is better, and 0.50 is no better than a coin flip.

Model	Inputs	Training AUC	Test AUC

Model	Inputs	Training AUC	Test AUC
1	age_over_75, prior_admissions	0.71	0.70
2	Model 1 plus 6 other clinical measures	0.78	0.76
3	Model 2 plus 40 further variables	0.95	0.66

Part C. Cost and threshold (29 points)

A test set has 1,000 discharged patients. 150 of them were readmitted. “Flag” means the model recommends a call. The table shows what happened at three thresholds.

Threshold	Patients flagged	Flagged and readmitted	Not flagged and readmitted	Flagged, not readmitted	Not flagged, not readmitted
0.50	60	45	105	15	835
0.25	200	85	65	115	735
0.10	370	120	30	250	600

1. (5 points) At threshold 0.50 the model flags 60 patients and 45 of them were readmitted (see the table). Compute sensitivity (recall) and precision, and say in one sentence what each means for the hospital.
2. (7 points) The first table below gives the total cost per 1,000 patients of four policies; the model at 0.10 is missing. (a) Compute it from the counts in the table above, using 9,000 per missed readmission and 1,000 per unnecessary call. (b) Which policy would you choose, and why? (c) The model at 0.50 is 88.0% accurate and calling nobody is 85.0% accurate. What does that comparison hide? (d) The hospital discharges 4,000 patients a year and running the model costs 400,000 a year. Taking the model at 0.10 against the best baseline, is it worth running, and at what running cost would you stop?
3. (5 points) (a) Compute the break-even probability. (b) From 0.50 to 0.10 accuracy falls from 88.0% to 72.0%, yet the cost falls (item 2). Sensitivity rises from 0.30 to 0.80 and specificity falls from 0.98 to 0.71. Explain why cost and accuracy move in different directions.
4. (7 points) The call team can make at most 200 calls per 1,000 patients. The CFO says a missed readmission may cost anywhere from 4,000 to 19,000. Use the second table below. (a) Which thresholds are feasible? (b) Recommend one threshold in two sentences and name the mistake you can afford. (c) At a missed-readmission cost of 19,000, how much would the hospital gain per 1,000 patients if it could make 370 calls? Is that worth asking for?
5. (5 points) Figure 1 below is the ROC curve of the model. Each point on the curve is one threshold; A, B and C are the three thresholds in the table above. (a) Match 0.50, 0.25 and 0.10 to A, B and C. (b) What does the dashed diagonal show? The area under the curve is 0.83: say in one sentence what that number means. (c) The break-even probability is 0.10. Which point do you choose, and why can the ROC curve alone not tell you the best point?

Table for item 2. Total cost per 1,000 patients under four policies (9,000 per missed readmission, 1,000 per unnecessary call).

Policy	Total cost
Call nobody	1,350,000
Call everybody	850,000
Model at threshold 0.50	960,000
Model at threshold 0.10	(to compute)

Table for item 4. Total cost per 1,000 patients, by what a missed readmission really costs.

Threshold	Patients flagged	Cost at 4,000	Cost at 19,000
0.50	60	435,000	2,010,000

Threshold	Patients flagged	Cost at 4,000	Cost at 19,000
0.25	200	375,000	1,350,000
0.10	370	370,000	820,000

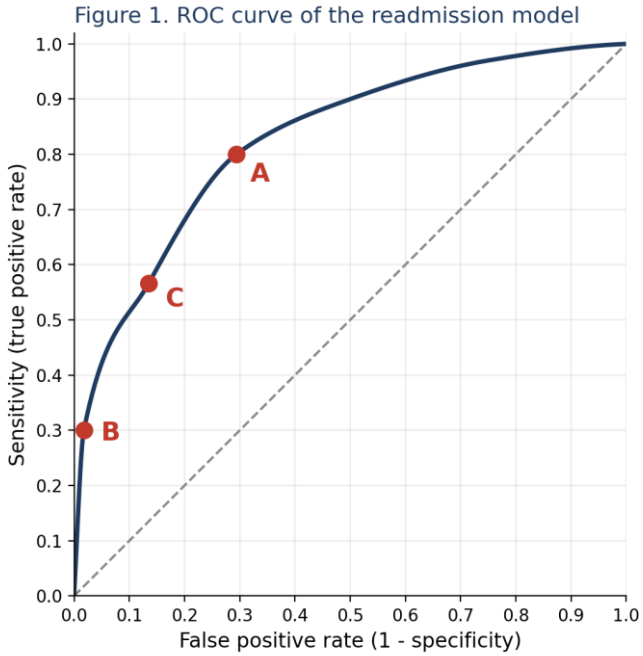


Figure 1. ROC curve of the readmission model, with three thresholds marked A, B and C

Part D. Audit the AI and decide (25 points)

An AI assistant wrote a summary of the readmission project for the hospital board. You are the reviewer.

Item 1 (10 points, 2 each). For each claim, write Supported or Not supported, and give one reason. Use Part B and Part C as your evidence.

- Claim 1: "The regression has $R^2 = 0.62$, so it will explain 62% of the variation in next year's readmissions."
- Claim 2: "Pharmacy visits have $p < 0.001$, so giving patients more pharmacy visits will reduce readmissions."
- Claim 3: "The model is 91% accurate on the test set, so it is ready to use." (The test readmission rate is 9%.)
- Claim 4: "At threshold 0.10, the model flags 120 of the 150 patients who were later readmitted."
- Claim 5: "Patients who joined the hospital's discharge-planning class were readmitted far less, so the class cuts readmissions."

Item 2 (7 points). At threshold 0.10, the notebook prints expected and actual cost for two groups.

Group	Patients	Expected cost	Actual cost
New to the hospital	300	120,000	270,000
Returning patients	700	230,000	250,000

(a) Which group do you trust less, and why? (b) Name one check you would run. (c) What result would make you change the threshold, and in which direction?

Item 3 (8 points). Write the JUDGMENT and the ACTION blocks of a board memo. Use at most 40 words each. They must agree with your numbers and claim no more than the evidence shows. The ACTION must also name one number you will monitor after launch.